

الجمهورية الجزائرية الديمقراطية الشعبية

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République Algérienne Démocratique et Populaire

وزارة التعليم العالي والبحث العلمي

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Ministère de l'Enseignement Supérieur et de la Recherche Scientifique



ECOLE NATIONALE
SUPÉRIEURE
D'INFORMATIQUE

المدرسة الوطنية العليا للإعلام الآلي

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École nationale Supérieure d'Informatique

SSR, SIQ2

LAB6A: ACTIVE SNIFFING

Done by:

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May 2026.

2. Perform A DHCP Starvation Attack Using Yersinia:

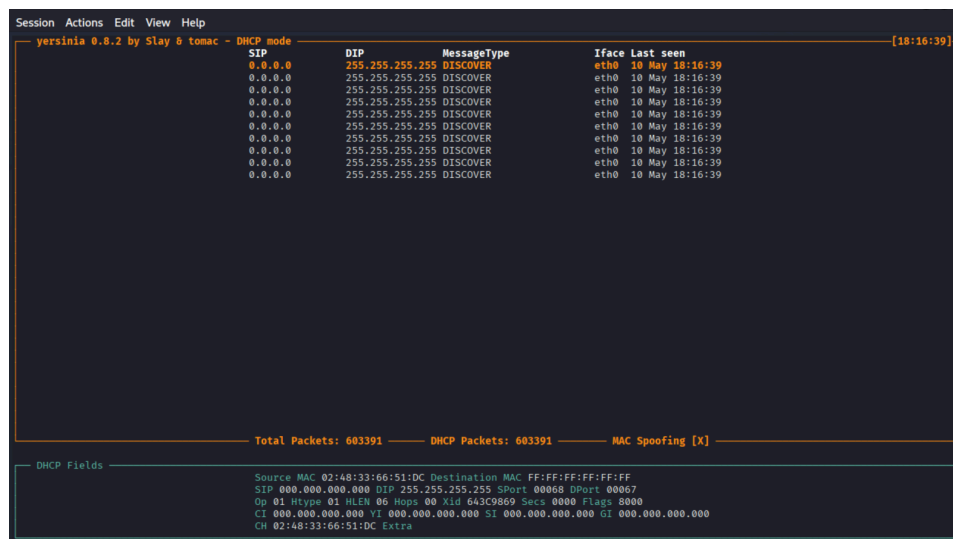
In the second task, we performed a DHCP starvation attack. The goal of this attack is to consume all available IP addresses from the DHCP server, preventing legitimate users from receiving an IP address.

We started Wireshark to capture traffic, then opened a terminal in Parrot Security and launched **Yersinia** in interactive mode:

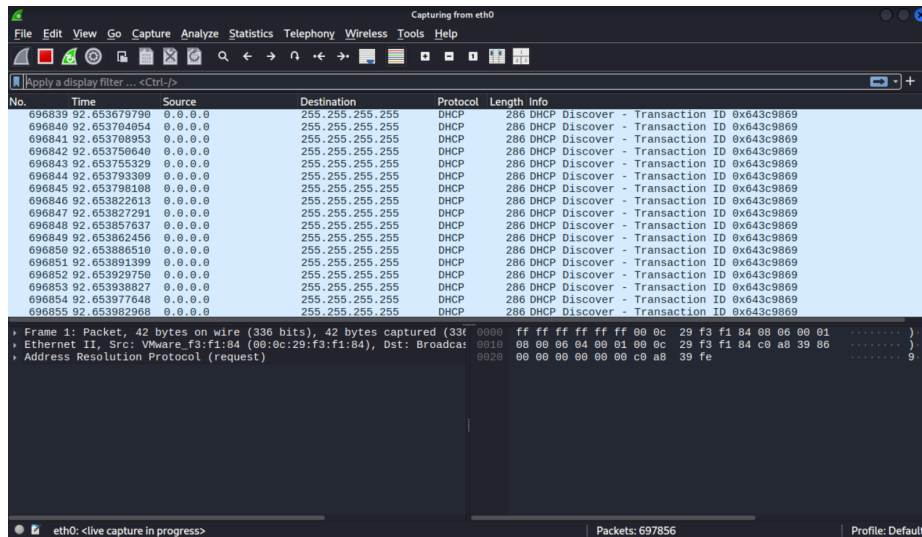
sudo su

yersinia -I

Inside Yersinia, we selected DHCP mode and launched the starvation attack.



During the attack, Yersinia generated many DHCP requests. In Wireshark, we observed a large number of DHCP packets being transmitted. This confirmed that the DHCP server was being flooded with requests.



3. Detect Network Sniffing:

3.1. ARP Poisoning Attack:

The third task focused on ARP poisoning using **Cain & Abel** on the Windows Server machine.

First, we configured Cain & Abel to scan the local network and identify active hosts.

IP address	MAC address	OUI fingerprint	Host name	B31	B16	B8	Gr	M0	M1	M3
192.168.57.1	005056C00008	VMware, Inc.		*	*	*	*	*	*	*
192.168.57.2	005056EA08BD	VMware, Inc.		*	*	*	*	*	*	*
192.168.57.254	005056FD6CAE	VMware, Inc.		*	*	*	*	*	*	*
192.168.57.130	000C298926C5	VMware, Inc.							*	
192.168.57.133	000C2945F13D	VMware, Inc.							*	

Then we selected two target systems: the Windows machine and the Parrot Security machine.

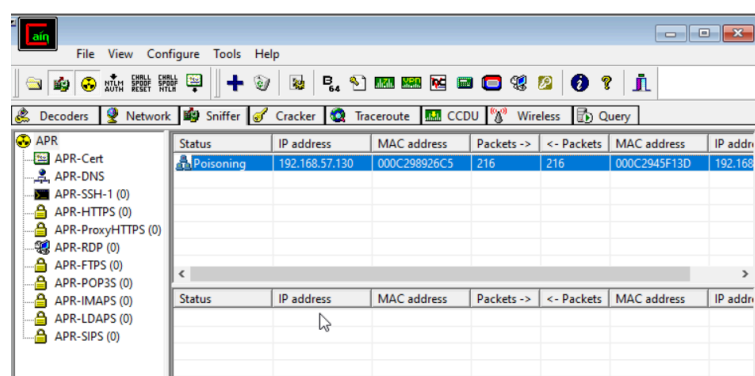
APR	Status	IP address	MAC address	Packets ->	<- Packets	MAC address	IP address
APR-Cert							
APR-DNS							
APR-SSH-1 (0)							
APR-HTTPS (0)							
APR-ProxyHTTPS (0)							
APR-Poisoning		192.168.57.130	000C298926C5	0	0	000C2945F13D	192.168.57.133

After selecting the targets, we started ARP poisoning between them. To generate network traffic between both systems, we used the following command on Parrot Security:

```
[root@parrot]~[/home/eden]
#hping3 192.168.57.130 -c 100000
HPING 192.168.57.130 (ens33 192.168.57.130): NO FLAGS are set, 40 headers + 0 data bytes
len=46 ip=192.168.57.130 ttl=128 DF id=0 sport=0 flags=RA seq=0 win=0 rtt=0.9 ms
len=46 ip=192.168.57.130 ttl=128 DF id=1 sport=0 flags=RA seq=1 win=0 rtt=1.7 ms
len=46 ip=192.168.57.130 ttl=128 DF id=2 sport=0 flags=RA seq=2 win=0 rtt=1.4 ms
len=46 ip=192.168.57.130 ttl=128 DF id=3 sport=0 flags=RA seq=3 win=0 rtt=2.1 ms
len=46 ip=192.168.57.130 ttl=128 DF id=4 sport=0 flags=RA seq=4 win=0 rtt=1.3 ms
len=46 ip=192.168.57.130 ttl=128 DF id=5 sport=0 flags=RA seq=5 win=0 rtt=2.0 ms
len=46 ip=192.168.57.130 ttl=128 DF id=6 sport=0 flags=RA seq=6 win=0 rtt=1.7 ms
len=46 ip=192.168.57.130 ttl=128 DF id=7 sport=0 flags=RA seq=7 win=0 rtt=1.4 ms
len=46 ip=192.168.57.130 ttl=128 DF id=8 sport=0 flags=RA seq=8 win=0 rtt=1.4 ms
len=46 ip=192.168.57.130 ttl=128 DF id=9 sport=0 flags=RA seq=9 win=0 rtt=1.0 ms
len=46 ip=192.168.57.130 ttl=128 DF id=10 sport=0 flags=RA seq=10 win=0 rtt=1.9 ms
len=46 ip=192.168.57.130 ttl=128 DF id=11 sport=0 flags=RA seq=11 win=0 rtt=1.7 ms
len=46 ip=192.168.57.130 ttl=128 DF id=12 sport=0 flags=RA seq=12 win=0 rtt=2.3 ms
len=46 ip=192.168.57.130 ttl=128 DF id=13 sport=0 flags=RA seq=13 win=0 rtt=2.0 ms
len=46 ip=192.168.57.130 ttl=128 DF id=14 sport=0 flags=RA seq=14 win=0 rtt=1.5 ms
len=46 ip=192.168.57.130 ttl=128 DF id=15 sport=0 flags=RA seq=15 win=0 rtt=2.3 ms
len=46 ip=192.168.57.130 ttl=128 DF id=16 sport=0 flags=RA seq=16 win=0 rtt=1.9 ms
len=46 ip=192.168.57.130 ttl=128 DF id=17 sport=0 flags=RA seq=17 win=0 rtt=1.6 ms
len=46 ip=192.168.57.130 ttl=128 DF id=18 sport=0 flags=RA seq=18 win=0 rtt=1.4 ms
len=46 ip=192.168.57.130 ttl=128 DF id=19 sport=0 flags=RA seq=19 win=0 rtt=1.1 ms
len=46 ip=192.168.57.130 ttl=128 DF id=20 sport=0 flags=RA seq=20 win=0 rtt=1.0 ms
```

This generated a large number of packets toward the target machine.

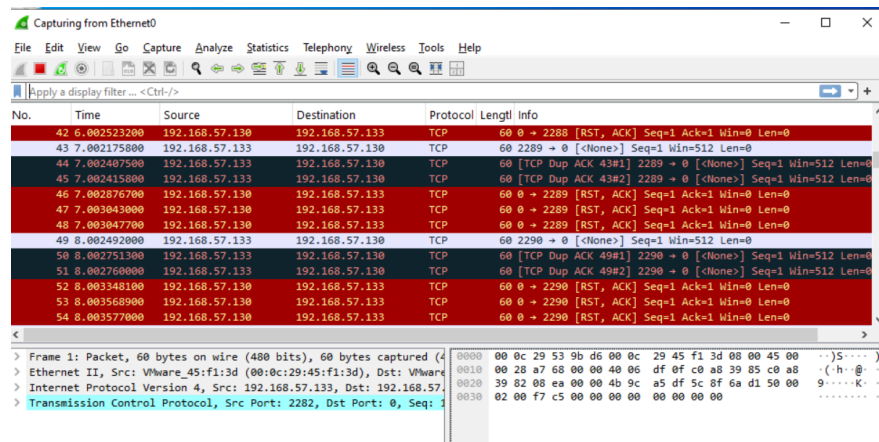
Cain & Abel successfully intercepted the communication between both systems, demonstrating a **Man-in-the-Middle (MITM) attack** where the attacker can monitor or modify traffic.



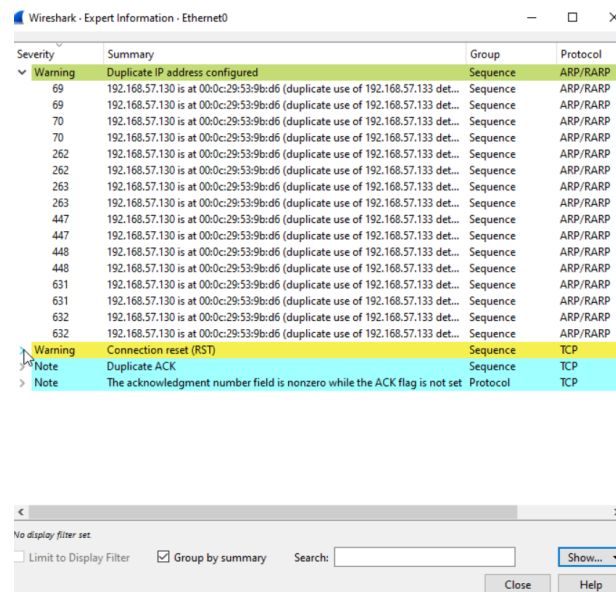
3.2. ARP Poisoning Detection:

After launching the ARP poisoning attack, we used Wireshark on the Windows Server machine to detect suspicious activity.

We enabled ARP/RARP detection settings in Wireshark and started packet capture on the network adapter.



After analyzing the captured packets using Expert Information, Wireshark displayed duplicate IP address warnings under the ARP/RARP protocol. This indicated that ARP poisoning was taking place on the network.



The severity level shown by Wireshark was Warning.

265	41.013309280	192.168.57.133	192.168.57.130	TCP	60 [TCP Dup ACK 2642] 2323 → 0 [RST, Seq=1 Win=512 Len=0]
267	41.013632780	192.168.57.130	192.168.57.133	TCP	60 0 → 2323 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
268	41.013751700	192.168.57.130	192.168.57.133	TCP	60 0 → 2323 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
269	41.013763480	192.168.57.130	192.168.57.133	TCP	60 0 → 2323 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
270	42.013484900	192.168.57.133	192.168.57.130	TCP	60 2324 → 0 [RST, Seq=1 Win=512 Len=0]
271	42.013774380	192.168.57.133	192.168.57.130	TCP	60 [TCP Dup ACK 27041] 2324 → 0 [RST, Seq=1 Win=512 Len=0]

>	Frame 263: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{6B001000-0000-0000-0000-000000000000}	0000	00 0c 29 45 f1 3d 00 0c 29
>	Ethernet II, Src: VMware_53:9b:d6 (00:0c:29:53:9b:d6), Dst: VMware_45:f1:3d (00:0c:29:45:f1:3d)	0010	00 00 00 04 00 02 00 0c 29
>	Address Resolution Protocol (reply)	0020	00 0c 29 45 f1 3d c0 a8 39
>	[Duplicate IP address detected for 192.168.57.130 (00:0c:29:53:9b:d6) - also in use by 00:0c:29:89:26:c5 [Frame showing earlier use of IP address: 261]		
>	[Seconds since earlier frame seen: 0]		
>	[Duplicate IP address detected for 192.168.57.133 (00:0c:29:45:f1:3d) - also in use by 00:0c:29:53:9b:d6]		

3.3. Detecting Promiscuous Mode:

In the final task, we checked whether a system on the network was running in promiscuous mode.

Using the Ubuntu machine, we opened a terminal and ran:

sudo su

nmap --script=sniffer-detect 192.168.57.132 ; IP_ATTACKER

This command used an Nmap NSE script to detect sniffing behavior.

The scan result indicated that the target system was likely in promiscuous mode, meaning its network interface was capable of capturing all traffic on the network segment.

```
[root@parrot]~/home/eden
#nmap --script=sniffer-detect 192.168.57.132
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-09 00:51 CET
Nmap scan report for 192.168.57.132
Host is up (0.00072s latency).
Not shown: 981 closed tcp ports (reset)
PORT      STATE SERVICE
53/tcp    open  domain
88/tcp    open  kerberos-sec
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
389/tcp   open  ldap
443/tcp   open  https
445/tcp   open  microsoft-ds
464/tcp   open  kpasswd5
593/tcp   open  http-ipc-epmap
636/tcp   open  ldapssl
990/tcp   open  ftps
993/tcp   open  imaps
995/tcp   open  pop3s
3268/tcp  open  globalcatLDAP
3269/tcp  open  globalcatLDAPssl
5061/tcp  open  sip-tls
5357/tcp  open  wsapi
5985/tcp  open  wsman
8080/tcp  open  http-proxy
MAC Address: 00:0C:29:53:9B:D6 (VMware)

Host script results:
|_sniffer-detect: Likely in promiscuous mode (tests: "1111111")
Nmap done: 1 IP address (1 host up) scanned in 2.51 seconds
```